Finite-Scale Notes on the Kakeya Needle Problem and the Besicovitch Set

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Abstract

We record finite-scale measurements for the Kakeya problem: a family of thin tubes around line segments pointing in every direction, inside the unit cube. As the tube radius δ decreases and the number of directions N increases, the volume of the union stays bounded well below the trivial sum, while the maximal overlap (multiplicity) grows with N. These are finite-scale observations only, not a proof of any theorem.

1 The construction

Let $\delta = 2^{-d}$ and take N directions distributed evenly on the upper hemisphere. Each tube has radius δ around a unit segment. A packing parameter $p \in [0, 1]$ slides the tubes toward the cube centre: $p = 0$ keeps them separated, $p = 1$ makes them sticky.

2 Measured quantities

For each configuration we measure the volume of the union $|\bigcup T|$ and the maximal multiplicity m by voxel counting at a finite resolution. The key observation is that doubling N increases the multiplicity while the union volume grows only slowly — consistent with the Besicovitch intuition that a measure-zero set can contain a unit segment in every direction.

3 Discussion

All numbers are finite-scale voxel measurements; the limit statements of the Kakeya–Besicovitch theory are not proved here. Increasing the packing parameter decreases the union volume toward the sticky-cluster limit.